

Enterprise High Availability Architecture using SQL Server Always On Availability Groups for Denim Manufacturing Industry

Project Overview

This project implements a **high availability and disaster recovery solution** using SQL Server Always On Availability Groups (AG) for a **Garment Manufacturing (Denim) enterprise system**.

The solution ensures:

- Continuous database availability
- Automatic failover
- Data redundancy across nodes
- Zero/minimal downtime for production systems

This architecture is designed to support **critical manufacturing operations**, including:

- Sales Orders
- Production Orders
- Cutting & Bundling
- Stitching Operations
- Worker & Machine Tracking



Problem Statement

In denim manufacturing environments:

- Production runs 24/7
- System downtime causes **financial loss and production delays**
- Single SQL Server instances create **single point of failure**
- Reporting and operational load affects performance

A **highly available, fault-tolerant system** is required.

Solution

Implemented:

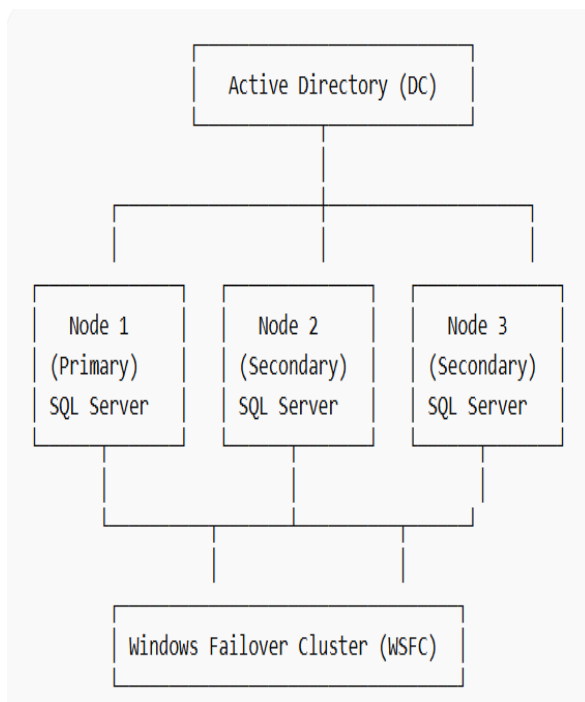
Windows Server Failover Cluster (WSFC)

SQL Server Always On Availability Groups

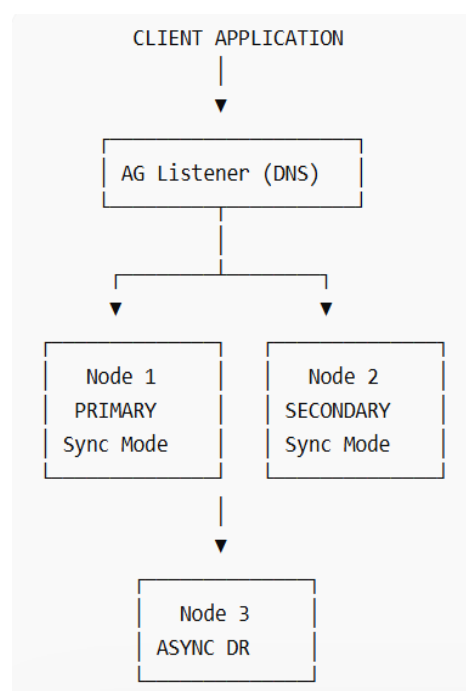
Multi-node synchronous & asynchronous replication

System Architecture

Physical Architecture



Availability Group Architecture



Industry Context – Denim Manufacturing

The system supports:

Core Modules:

- Sales Order Management
- Production Planning
- Cutting & Bundling
- Stitching Operations
- Quality Control
- Packing & Shipment

Database Design Overview

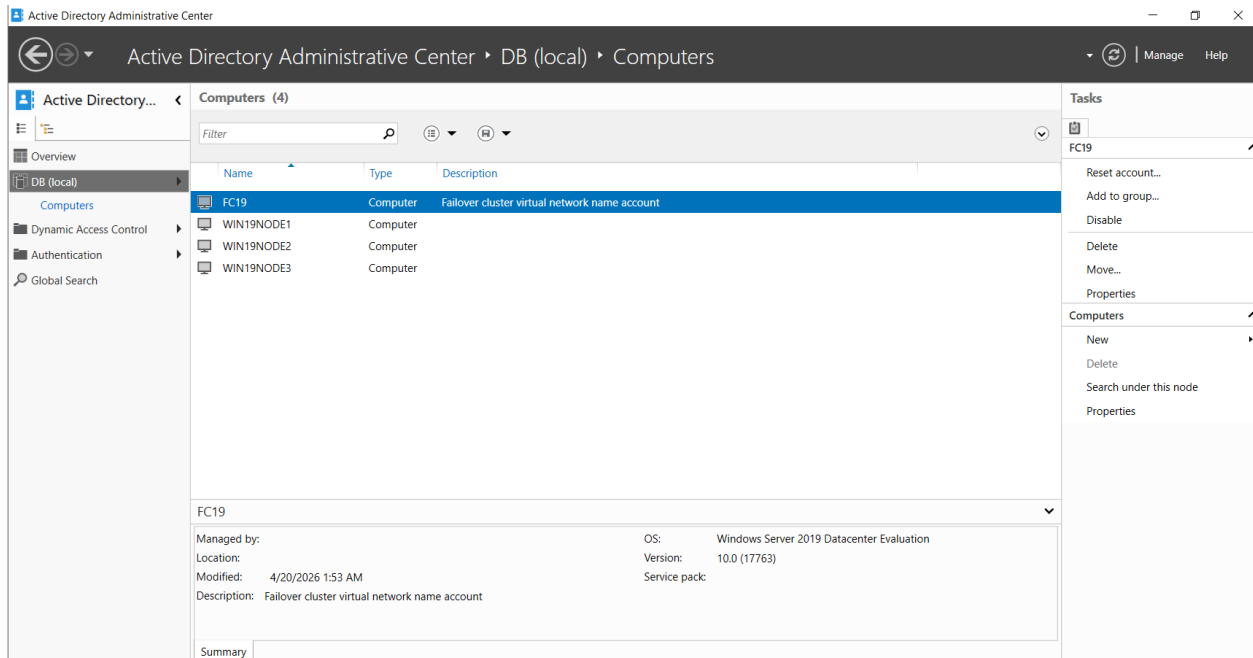
Key Databases:

Database Name	Purpose
DenimCoreDB	OLTP (Production System)
DenimReportingDB	Reporting
MSDB / System DBs	SQL Jobs & AG

Implementation Steps

Step 1: Active Directory Setup

- Domain Controller configured (DB)
- All nodes joined to domain

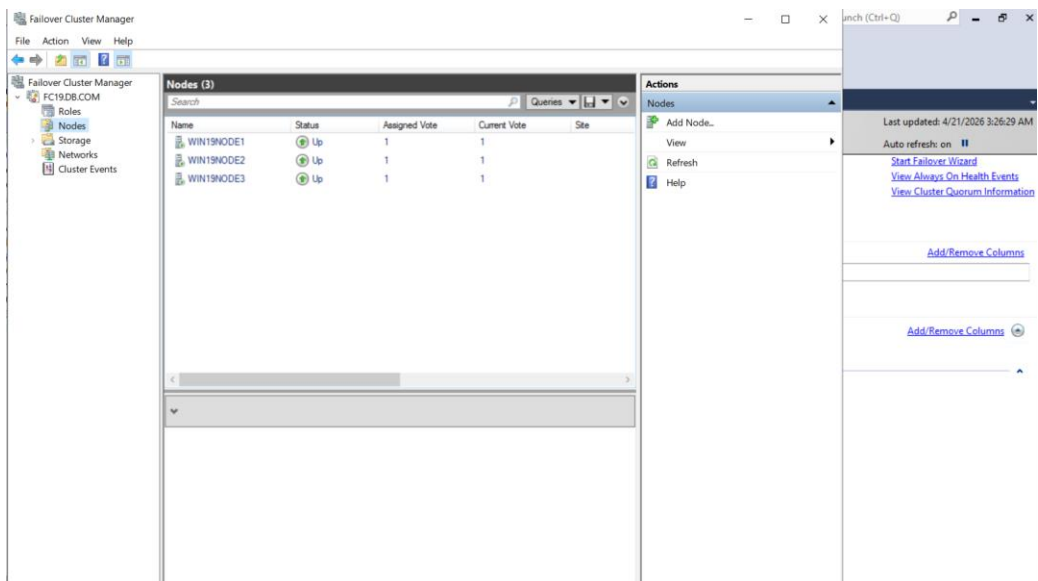


Step 2: Windows Failover Cluster (WSFC)

- Installed Failover Clustering feature
- Validation tests executed
- Cluster created

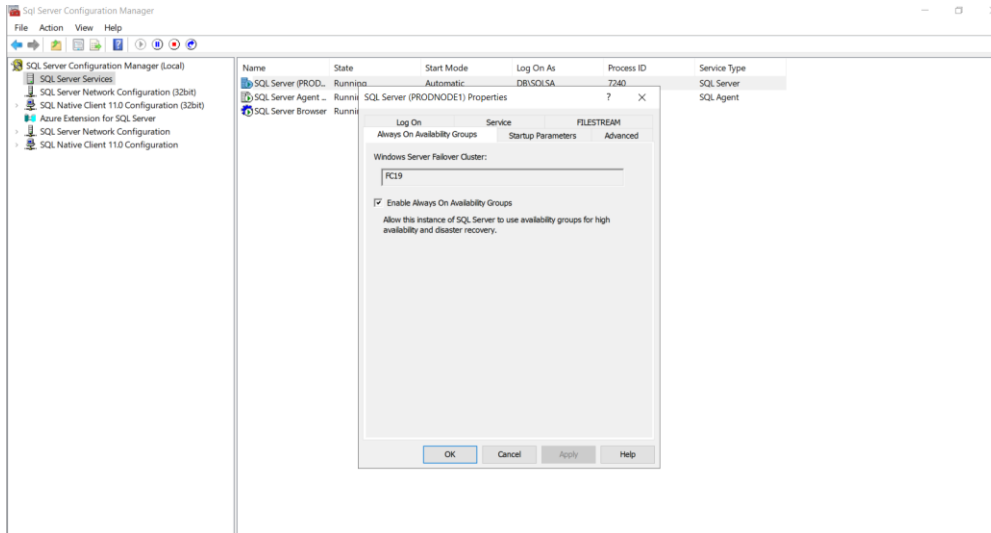
Cluster Name: FC19

Nodes: WIN19Node1, WIN19Node2, WIN19Node3



Step 3: SQL Server Configuration

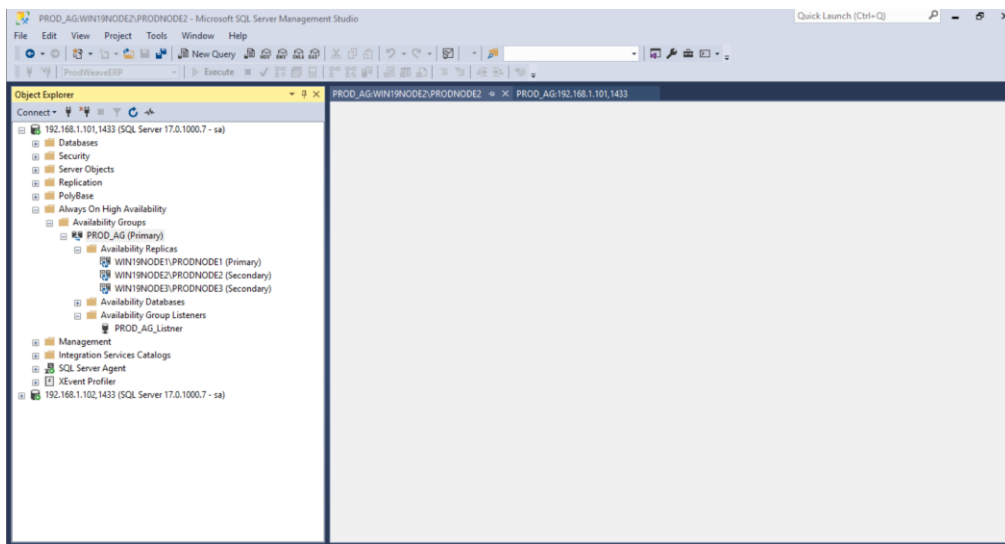
- Installed SQL Server on all nodes
- Enabled Always On feature



Step 4: Database Preparation

- Full backup
- Transaction log backup

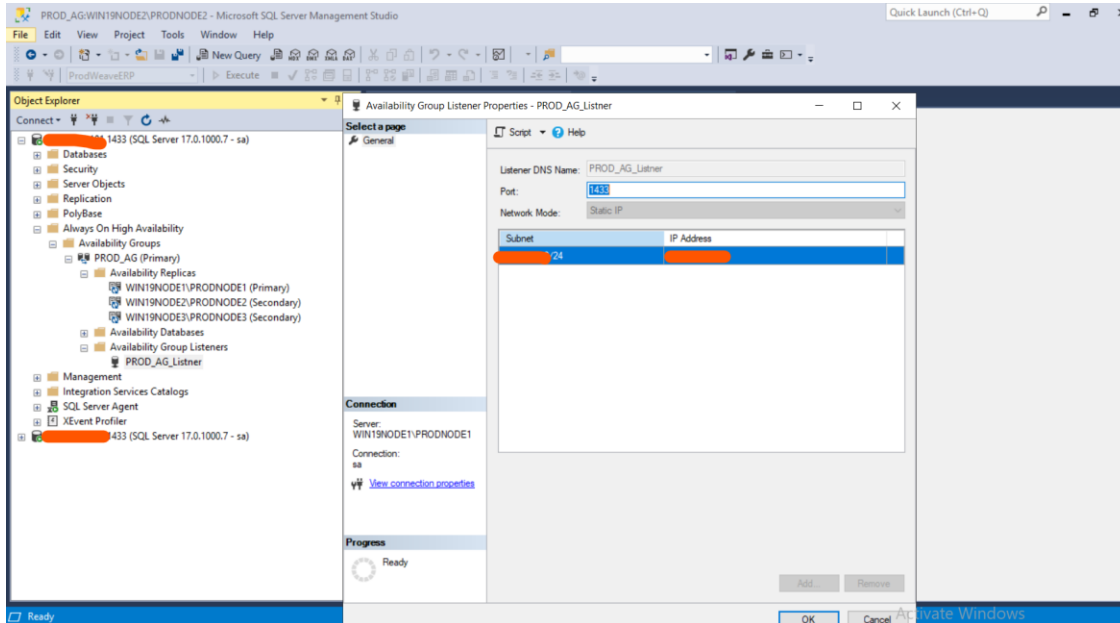
Step 5: Availability Group Creation



Step 6: Listener Configuration

Listener Name: PROD_AG_Listner

Port: 1433



Data Flow

Application → AG Listener → Primary Node

Primary Node:

Write Operations

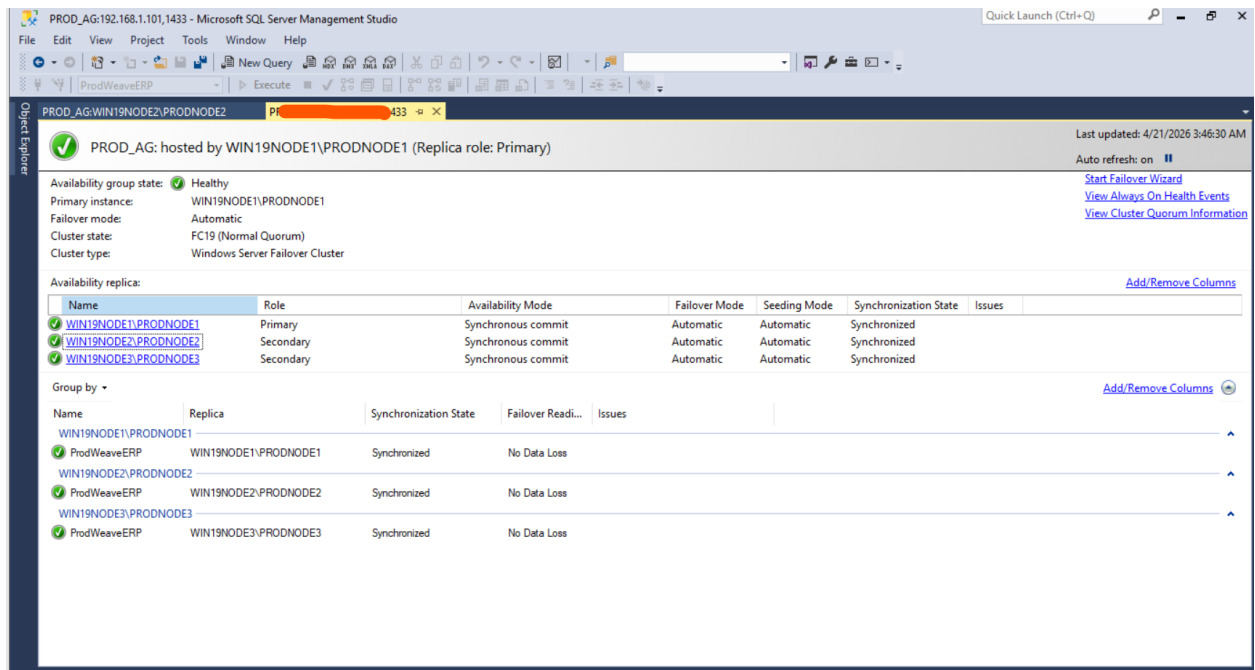
↓ (Transaction Log)

Secondary Node 2:

Synchronous Replica (Real-time)

Secondary Node 3:

Asynchronous Replica (DR Site)



Security

- Domain-based authentication
- Encrypted endpoints
- Restricted access on secondary replicas
- Read-only routing configured

Always on Availability Groups Features

1. Automatic Failover Node1 <-> Node2
2. Manual Failover Node1 -> Node3
3. Zero Data Loss (Synchronous Mode)
4. Disaster Recovery (Asynchronous Model)
5. Load Distribution (Read-only Secondary)

Denim Industry Use Case

1. Production Continuity with zero downtime

2. Real Time Tracking
3. Reporting Isolation

Challenges & Solutions

1. Cluster Validation Failures

Solution: Fixed network

2. End Points Communication

Solution: Enabled TCP port 5022

3. Permission Issue for services

Solution: Dedicated domain user

Outcome

- Achieved **99.99% availability**
- Zero downtime during failover tests
- Improved production efficiency
- Scalable enterprise architecture

Technologies Used

- Microsoft SQL Server
- Windows Server Failover Clustering
- Always On Availability Groups
- T-SQL

Note: All the configuration scripts are attached in sql file for review.